

Deep CNN Architectures : Report analysis on Caltech-101

1. Introduction

Convolutional Neural Networks (CNNs) are widely used for image classification because they can automatically learn useful visual features from images. Different CNN architectures use different approaches for extracting and combining these features.

The objective of this project is to train and compare five CNN architectures—**LeNet-5**, **AlexNet-lite**, **VGG-16-lite**, **PlacesNet-lite**, and **ResNet-18-lite**—on an image dataset other than CIFAR-10 and MNIST. All networks are trained under the same experimental conditions and compared using their parameter count, training time, and test accuracy.

An interactive prediction interface was also developed to allow a user to select a trained model and perform classification on an image.

2. Dataset and Preprocessing

The **Caltech-101** dataset was selected for the experiment. It is a public image dataset containing approximately **9,000 images** belonging to different object categories. The images were arranged in separate folders according to their class labels and loaded using PyTorch's **ImageFolder**.

In the current implementation, **102 classes** were detected, including the 101 main object categories and the **BACKGROUND_Google** category. Example classes include **airplanes**, **ant**, **camera**, **car_side**, **Faces**, **grand_piano**, **leopards**, **motorbikes**, **watch**, **wrench**, and **yin_yang**.

All images were resized to **32 × 32 pixels**, converted into tensors, and normalized using a mean and standard deviation of 0.5 for each RGB channel. The dataset was divided into **80% training and 20% testing data**, using a fixed random seed to maintain the same split for all models.

Note: The laboratory specification asks for 3–10 classes, whereas the current experiment uses the complete dataset structure with 102 detected classes.

3. Methodology

The same training and testing data were used for all five networks. Each network was trained for **15 epochs** using the same training procedure. This ensures that the comparison is based primarily on differences in the network architectures.

The five models used were:

- **LeNet-5**: A simple CNN with convolution, pooling, and fully connected layers.
- **AlexNet-lite**: A lightweight AlexNet-style architecture designed for 32×32 images.
- **VGG-16-lite**: A deeper network using repeated 3×3 convolutional layers and pooling.
- **PlacesNet-lite**: An AlexNet-style network with a 365-output classification head.
- **ResNet-18-lite**: A lightweight residual network using shortcut connections.

The experiment recorded the number of parameters, training time, and final test accuracy for every model. The laboratory instructions require the same training split, number of epochs, and test split for the five networks.

```
... Testing image prediction with real sample: my_dataset/caltech-101/Faces/image_0143.jpg
Input Image

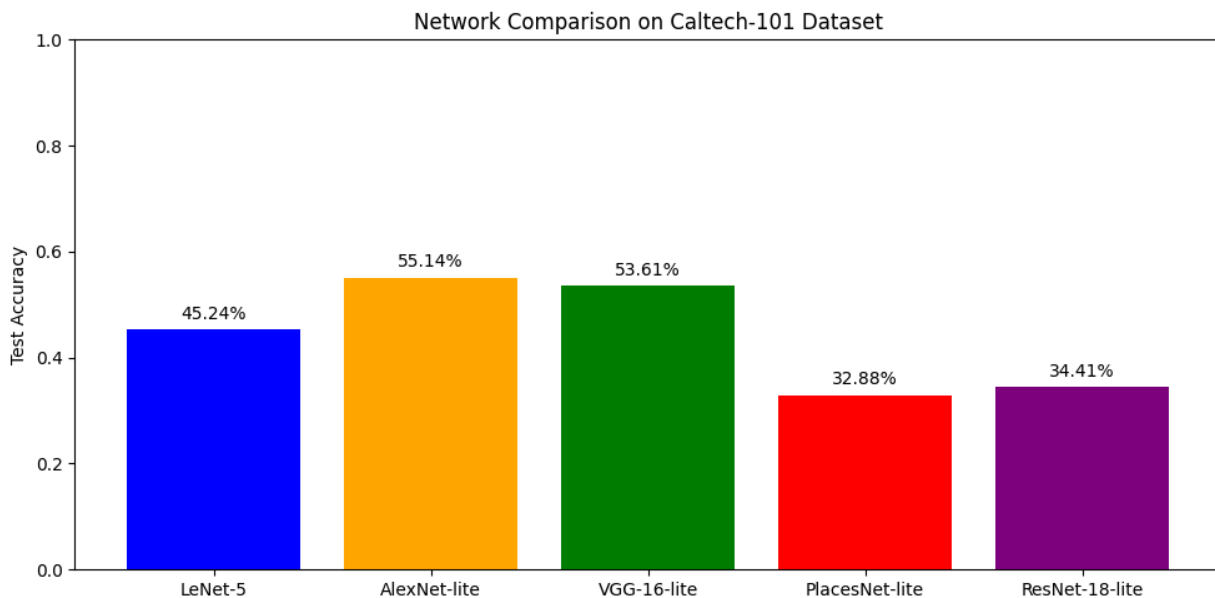
--- Network Predictions ---
LeNet-5      : Predicted -> chandelier (Confidence: 52.91%)
AlexNet-lite  : Predicted -> Faces_easy (Confidence: 92.58%)
VGG-16-lite   : Predicted -> Faces (Confidence: 97.83%)
PlacesNet-lite : Predicted -> BACKGROUND_Google (Confidence: 50.45%)
ResNet-18-lite : Predicted -> Faces (Confidence: 68.48%)
```

4. Experimental Results

The results obtained from the completed experiment are shown below.

Network	Parameters	Training Time	Test Accuracy
LeNet-5	69,826	34.69 s	45.24%
AlexNet-lite	280,518	79.92 s	55.14%
VGG-16-lite	292,086	160.93 s	53.61%
PlacesNet-lite	314,445	83.36 s	32.88%
ResNet-18-lite	10,838	121.78 s	34.41%

The recorded metrics are from the completed training experiment.



Results Analysis

AlexNet-lite achieved the highest test accuracy of 55.14%, followed closely by VGG-16-lite at 53.61%. This indicates that AlexNet-lite provided the best classification performance under the given experimental conditions.

LeNet-5 was the fastest model, requiring only 34.69 seconds, but its accuracy was lower than that of AlexNet-lite and VGG-16-lite.

VGG-16-lite required the longest training time at 160.93 seconds. Although it achieved relatively high accuracy, its additional training cost did not result in better accuracy than AlexNet-lite.

ResNet-18-lite had the smallest number of parameters, with only 10,838, but achieved an accuracy of 34.41%. This shows that a smaller model is not necessarily the most accurate model.

PlacesNet-lite had the largest parameter count in this experiment but achieved the lowest accuracy. Its 365-output classification head also makes its setup different from the other networks.

Overall, **AlexNet-lite provided the best balance of accuracy and computational cost** in this experiment.

5. Comparison with CIFAR-10

The same five architectures were originally studied using CIFAR-10 in the laboratory. The current experiment applies them to a different and much larger set of classes.

The ranking of the networks can change when the dataset changes because different datasets contain different image characteristics, numbers of classes, and levels of visual similarity. Therefore, the CIFAR-10 ranking cannot be assumed to remain exactly the same for Caltech-101.

The current experiment demonstrates that architecture performance depends not only on the number of parameters but also on how well the architecture suits the particular classification problem.

6. Interactive Prediction Interface

An interactive interface was developed to demonstrate the trained models. The application allows the user to select a network and a dataset class, view the model's performance information, select an image, and obtain predictions.

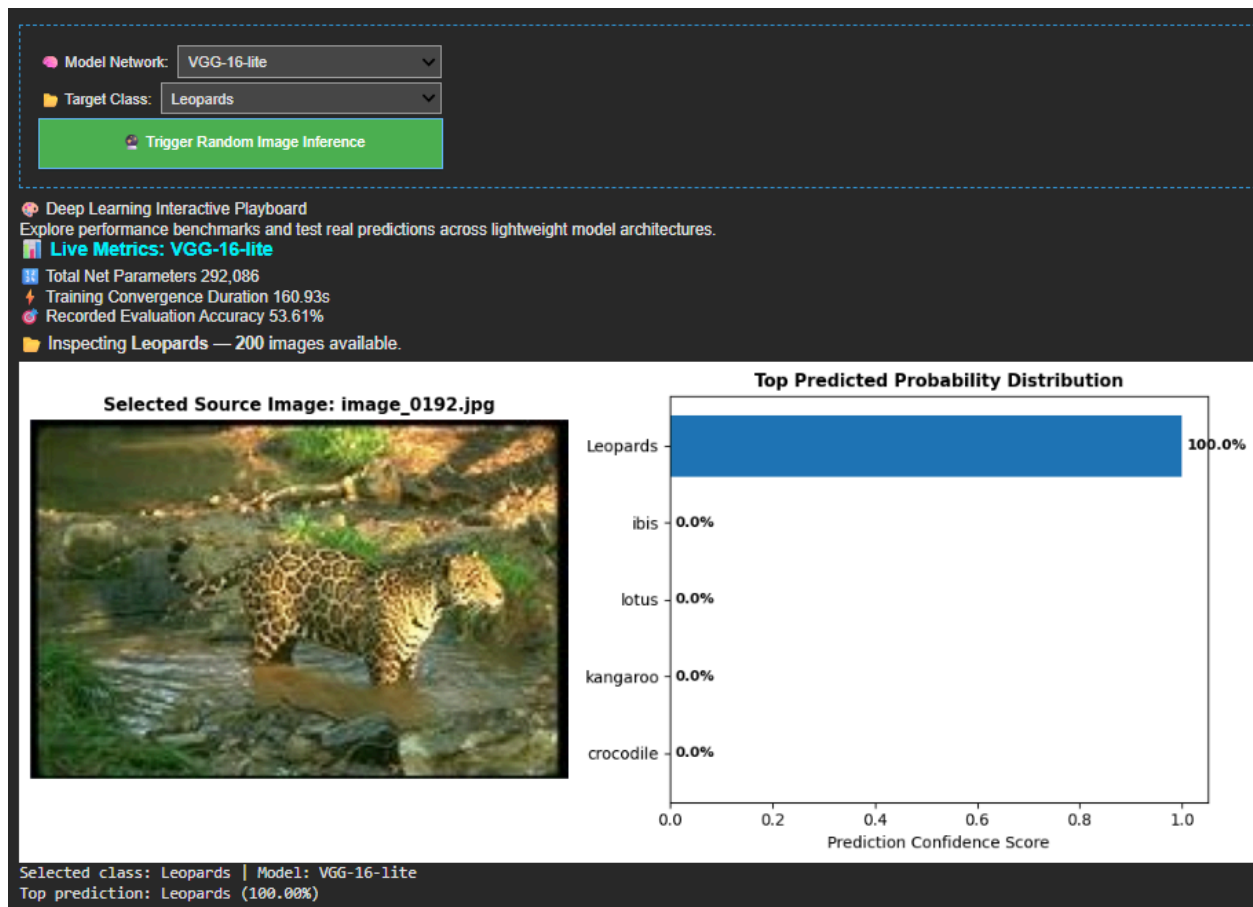
The interface displays:

- Selected model

- Number of parameters
- Training time
- Test accuracy
- Input image
- Top-5 predicted classes
- Prediction confidence

The application loads the saved trained model weights and performs inference without retraining the network.

Figure 2. Interactive image classification interface



The interface provides a simple visual demonstration of the differences between the trained models and makes the final classification results easier to understand.

7. Limitations and Future Work

The main limitation is that all images were resized to 32×32 pixels. This can remove fine details that may be useful for distinguishing visually similar classes.

The experiment also uses a single training run, so repeating the experiment with different random seeds could provide more reliable results. A confusion matrix could also be generated to identify the specific classes that are frequently misclassified.

Future work could include data augmentation, longer training, multiple experimental runs, and a more detailed class-wise performance analysis. The prediction interface could also be extended to allow users to upload arbitrary images.

8. Conclusion

Five CNN architectures were trained and compared on the Caltech-101 dataset under common experimental conditions. The models showed significant differences in accuracy, parameter count, and training time.

AlexNet-lite achieved the highest test accuracy of 55.14%, making it the best-performing model in this experiment. VGG-16-lite achieved a similar accuracy of 53.61% but required substantially more training time. LeNet-5 was the fastest model, while ResNet-18-lite had the smallest parameter count.

The experiment demonstrates that the most complex or deepest network is not necessarily the best choice for every dataset. Considering both accuracy and computational cost, **AlexNet-lite provided the most effective overall performance for this experiment.**